

Project: EduLink - School Education & Digital Learning Management System

1. Introduction

EduLink is a web-based platform designed for schools, education boards, and local governments to manage student enrollment, digital learning, examinations, and compliance. It enables students to access learning materials, teachers to manage classes, administrators to oversee operations, and regulators to monitor adherence to education policies.

The system supports workflows for student registration, course management, digital learning content delivery, examination scheduling, grading, compliance monitoring, and analytics. It ensures transparency and accountability by maintaining audit trails, dashboards, and performance metrics across school education operations.

Actors / Users

- **Student:** Registers, enrolls in courses, accesses digital learning, tracks grades.
- **Teacher:** Manages classes, uploads learning materials, conducts examinations.
- **School Administrator:** Oversees school operations, manages staff and students.
- **Education Board Officer:** Monitors compliance, manages programs, generates reports.
- **Compliance Officer:** Ensures adherence to education policies, audits records.
- **Regulator:** Reviews compliance reports, monitors accreditation.

2. Module Overview

- 2.1 Identity & Access Management
- 2.2 Student Registration & Profile Management
- 2.3 Course & Class Management
- 2.4 Digital Learning Content Delivery
- 2.5 Examination & Grading Management
- 2.6 Attendance & Performance Tracking
- 2.7 Compliance & Audit Management
- 2.8 Reporting & Analytics
- 2.9 Notifications & Alerts

3. Architecture Overview

- **Frontend:** Angular or React for responsive dashboards.
- **Backend:** REST API-based microservices (Identity, Students, Courses, Learning, Exams, Attendance, Compliance, Reports).

- **Database:** Relational DB (MySQL/PostgreSQL/SQL Server).
- **Deployment:** Cloud/on-prem with API gateway, WAF, centralized logging.

4. Module Wise Design

4.1 Identity & Access Management

Features:

Provides secure authentication, role-based access control, and audit logging.

Entities:

- User(UserID, Name, Role [Student/Teacher/Admin/Board/Compliance/Regulator], Email, Phone, Status)
- AuditLog(AuditID, UserID, Action, Resource, Timestamp)

4.2 Student Registration & Profile Management

Features:

Registers students, maintains profiles, validates documents.

Entities:

- Student(StudentID, Name, DOB, Gender, Address, ContactInfo, EnrollmentDate, Status)
- StudentDocument(DocumentID, StudentID, DocType [IDProof/ReportCard], FileURI, UploadedDate, VerificationStatus)

4.3 Course & Class Management

Features:

Creates courses, manages classes, assigns teachers, tracks delivery.

Entities:

- Course(CourseID, Title, Subject, GradeLevel, Credits, Status)
- Class(ClassID, CourseID, TeacherID, Schedule, Status)

4.4 Digital Learning Content Delivery

Features:

Uploads and delivers digital learning materials, tracks usage.

Entities:

- LearningMaterial(MaterialID, CourseID, Title, FileURI, UploadedDate, Status)

- Assignment(AssignmentID, CourseID, StudentID, Title, SubmissionDate, Status)

4.5 Examination & Grading Management

Features:

Conducts examinations, records grades, tracks performance.

Entities:

- Exam(ExamID, CourseID, Type [Midterm/Final], Date, Status)
- Grade(GradeID, ExamID, StudentID, Score, Grade, Status)

4.6 Attendance & Performance Tracking

Features:

Tracks student attendance, monitors performance metrics.

Entities:

- Attendance(AttendanceID, StudentID, ClassID, Date, Status)
- PerformanceMetric(MetricID, StudentID, CourseID, Score, Date, Status)

4.7 Compliance & Audit Management

Features:

Ensures adherence to education policies, audits records.

Entities:

- ComplianceRecord(ComplianceID, EntityID, Type [Course/Exam/Attendance], Result, Date, Notes)
- Audit(AuditID, OfficerID, Scope, Findings, Date, Status)

4.8 Reporting & Analytics

Features:

Generates dashboards and reports on enrollment, courses, exams, attendance, and compliance.

Entities:

- Report(ReportID, Scope [Student/Course/Exam], Metrics, GeneratedDate)

4.9 Notifications & Alerts

Features:

Sends alerts for enrollment deadlines, exam schedules, assignment submissions, compliance reminders.

Entities:

- Notification(NotificationID, UserID, EntityID, Message, Category [Enrollment/Course/Exam/Compliance], Status, CreatedDate)

5. Deployment Strategy

- **Local:** Angular/React frontend, Spring Boot/.NET Core backend, local DB.
- **Production:** Cloud/on-prem deployment, API gateway, WAF, centralized logging.

6. Database Design

Tables:

- User(UserID, Name, Role, Email, Phone, Status)
- AuditLog(AuditID, UserID, Action, Resource, Timestamp)
- Student(StudentID, Name, DOB, Gender, Address, ContactInfo, EnrollmentDate, Status)
- StudentDocument(DocumentID, StudentID, DocType, FileURI, UploadedDate, VerificationStatus)
- Course(CourseID, Title, Subject, GradeLevel, Credits, Status)
- Class(ClassID, CourseID, TeacherID, Schedule, Status)
- LearningMaterial(MaterialID, CourseID, Title, FileURI, UploadedDate, Status)
- Assignment(AssignmentID, CourseID, StudentID, Title, SubmissionDate, Status)
- Exam(ExamID, CourseID, Type, Date, Status)
- Grade(GradeID, ExamID, StudentID, Score, Grade, Status)
- Attendance(AttendanceID, StudentID, ClassID, Date, Status)
- PerformanceMetric(MetricID, StudentID, CourseID, Score, Date, Status)
- ComplianceRecord(ComplianceID, EntityID, Type, Result, Date, Notes)
- Audit(AuditID, OfficerID, Scope, Findings, Date, Status)
- Report(ReportID, Scope, Metrics, GeneratedDate)
- Notification(NotificationID, UserID, EntityID, Message, Category, Status, CreatedDate)

7. User Interface Design

- **Student Portal:** Enrollment, course access, digital learning, exam schedules.
- **Teacher Dashboard:** Class management, material uploads, grading.
- **School Administrator Console:** Student and staff management, compliance tracking, reporting.

- **Education Board Officer Workbench:** Program monitoring, compliance oversight, reporting.
- **Compliance Officer Console:** Policy monitoring, audits, compliance reporting.
- **Regulator Dashboard:** Accreditation monitoring, compliance review.

8. Non-Functional Requirements

- **Performance:** Handle 50,000 concurrent users across schools.
- **Security:** Role-based access, encrypted data storage, immutable audit logs.
- **Scalability:** Support multi-school, multi-board rollout.
- **Availability:** 99.9% uptime.
- **Maintainability:** Modular microservices, API versioning, automated migrations.
- **Observability:** Centralized logging, KPIs (exam pass rate, attendance percentage, compliance adherence).

9. Assumptions & Constraints

- Initial rollout for a single school district before multi-district expansion.
- Notifications limited to in-app and SMS/email alerts.
- Fully implementable using Java Spring Boot/ASP.NET Core, Angular/React, and relational DB.